

C Programming - Day 7 Practical Report

Program: B.Tech Computer Science and Engineering

Question 1: Hydraulic Lift Output Force

Objective: Write a C program to find hydraulic lift output force when input force, input area and output area are given.

Inputs

- Input Force
- Input Area
- Output Area

Formula

Output Force = (Input Force × Output Area) / Input Area

Test Case Table

Test Case	Input(s)	Expected Output	Actual Output	Result
1	100,20,100	500.00	500.00	Pass
2	50,10,50	250.00	250.00	Pass
3	80,40,120	240.00	240.00	Pass

Algorithm

1. Start
2. Read Input Force, Input Area and Output Area.
3. Calculate Output Force = (Input Force × Output Area) / Input Area.
4. Display Output Force.
5. Stop.

C Program

```
#include <stdio.h>

int main() {
    float inputForce,inputArea,outputArea,outputForce;
    printf("Enter Input Force: ");
    scanf("%f",&inputForce);
    printf("Enter Input Area: ");
    scanf("%f",&inputArea);
    printf("Enter Output Area: ");
```

```
scanf("%f",&outputArea);
outputForce=(inputForce*outputArea)/inputArea;
printf("Output Force = %.2f",outputForce);
return 0;
}
```

Compilation Screenshot:

```
(kali@kali)-[~/desktop]
$ gcc q1.c -o q1

(kali@kali)-[~/desktop]
$
```

Execution Screenshot:

```
(kali@kali)-[~/desktop]
$ ./q1
Enter Input Force: 50
Enter Input Area: 10
Enter Output Area: 50
Output Force = 250.00
```

Question 2: Impedance of an LCR Circuit

Objective: Write a C program to find impedance when resistance, inductive reactance and capacitive reactance are given.

Inputs

- Resistance
- Inductive Reactance
- Capacitive Reactance

Formula

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Test Case Table

Test Case	Input(s)	Expected Output	Actual Output	Result
-----------	----------	-----------------	---------------	--------

1	10,20,5	18.03	18.03	Pass
2	15,25,10	18.03	21.21	Pass
3	5,12,4	9.43	9.43	Pass

Algorithm

6. Start
7. Read Resistance, XL and XC.
8. Calculate $Z = \sqrt{R^2 + (XL - XC)^2}$.
9. Display impedance.
10. Stop.

C Program

```
#include <stdio.h>
#include <math.h>

int main() {
    float R,XL,XC,Z;
    printf("Enter Resistance: ");
    scanf("%f",&R);
    printf("Enter Inductive Reactance: ");
    scanf("%f",&XL);
    printf("Enter Capacitive Reactance: ");
    scanf("%f",&XC);
    Z=sqrt((R*R)+((XL-XC)*(XL-XC)));
    printf("Impedance = %.2f",Z);
    return 0;
}
```

Compilation Screenshot:

```
(kali@kali)-[~/desktop]
$ gcc q2.c -o q2 -lm
(kali@kali)-[~/desktop]
$
```

Execution Screenshot:

```
(kali㉿kali)-[~/desktop]
$ ./q2
Enter Resistance: 10
Enter Inductive Reactance: 20
Enter Capacitive Reactance: 5
Impedance = 18.03
```

Question 3: Work Done in Lifting a Body

Objective: Write a C program to find work done when mass, gravity and height are given.

Inputs

- Mass
- Gravity
- Height

Formula

Work = Mass $\times$ Gravity $\times$ Height

Test Case Table

Test Case	Input(s)	Expected Output	Actual Output	Result
1	10,9.8,5	490.00	490.00	Pass
2	20,9.8,2	392.00	392.00	Pass
3	15,10,4	600.00	600.00	Pass

Algorithm

11. Start
12. Read Mass, Gravity and Height.
13. Calculate Work = Mass $\times$ Gravity $\times$ Height.
14. Display Work Done.
15. Stop.

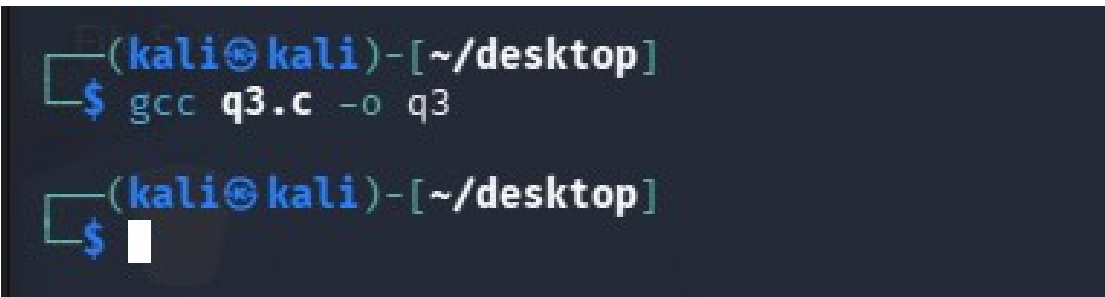
C Program

```
#include <stdio.h>

int main() {
    float mass,gravity,height,work;
    printf("Enter Mass: ");
```

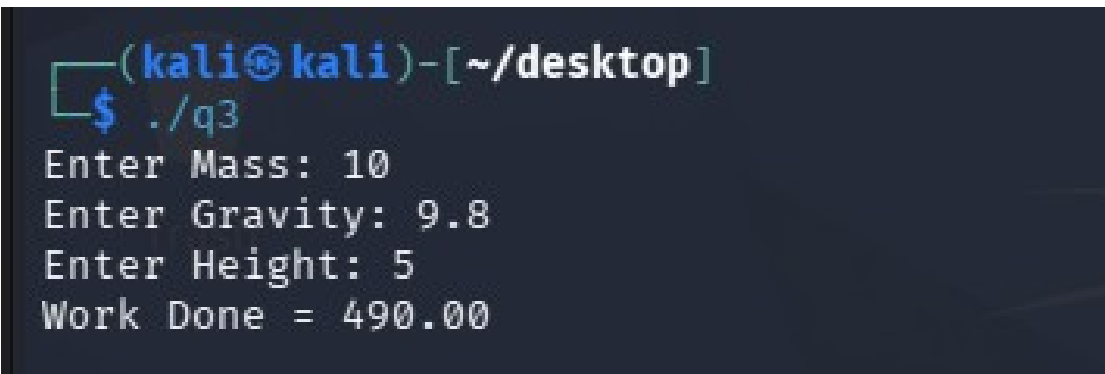
```
scanf("%f",&mass);  
printf("Enter Gravity: ");  
scanf("%f",&gravity);  
printf("Enter Height: ");  
scanf("%f",&height);  
work=mass*gravity*height;  
printf("Work Done = %.2f",work);  
return 0;  
}
```

Compilation Screenshot:



A terminal window with a dark background. The prompt is `(kali㉿kali)-[~/desktop]`. The user enters `$ gcc q3.c -o q3`. The prompt changes to `(kali㉿kali)-[~/desktop]` and the user enters `$` followed by a space.

Execution Screenshot:



A terminal window with a dark background. The prompt is `(kali㉿kali)-[~/desktop]`. The user enters `$ ./q3`. The program outputs: `Enter Mass: 10`, `Enter Gravity: 9.8`, `Enter Height: 5`, and `Work Done = 490.00`.